EXPERIMENT 6b: INTERACTIVE DASHBOARDS WITH TABLEAU

LEARNING OBJECTIVES:

- To understand interactive data visualization using Tableau.
- To load and explore the Sample Superstore dataset.
- To create bar, line, and scatter charts.
- To analyze sales and profit using different dimensions.
- To apply filters and dashboard actions.
- To design an interactive and user-friendly dashboard.
- To export or publish the completed Tableau dashboard.

METHODS / PYTHON LIBRARIES USED:

Library/method	Purpose	Reason for use
Tableau	Data visualization and dashboard creation	Creates interactive visualizations
Bar Chart	Display Sales by Category	Easy comparison of categories
Line Chart	Display Sales over time	Identifies sales trends
Dashboard Actions	Connect different charts	Makes the dashboard interactive
Containers	Arrange dashboard objects	Provides clean dashboard layout

PROGRAM CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Sample - Superstore.csv", encoding="latin1")
df["Order Date"] = pd.to_datetime(df["Order Date"])

print(df.head())
print("Shape:", df.shape)
print(df.info())
```

```
print(df.isnull().sum())

df.groupby("Category")["Sales"].sum().plot(kind="bar", title="Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()

sales = df.groupby([df["Order Date"].dt.to_period("M"),
"Region"])["Sales"].sum().reset_index()
sales["Order Date"] = sales["Order Date"].dt.to_timestamp()

for region, data in sales.groupby("Region"):
    plt.plot(data["Order Date"], data["Sales"], label=region)

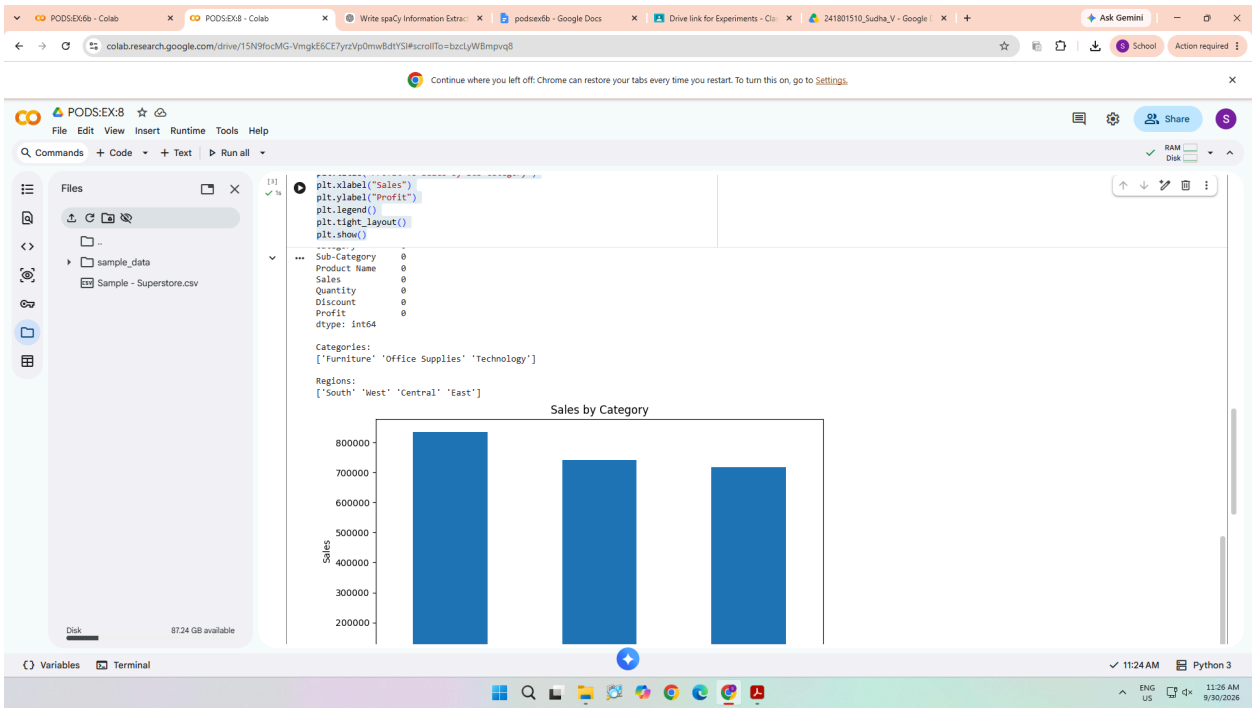
plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.legend()
plt.xticks(rotation=45)
plt.show()

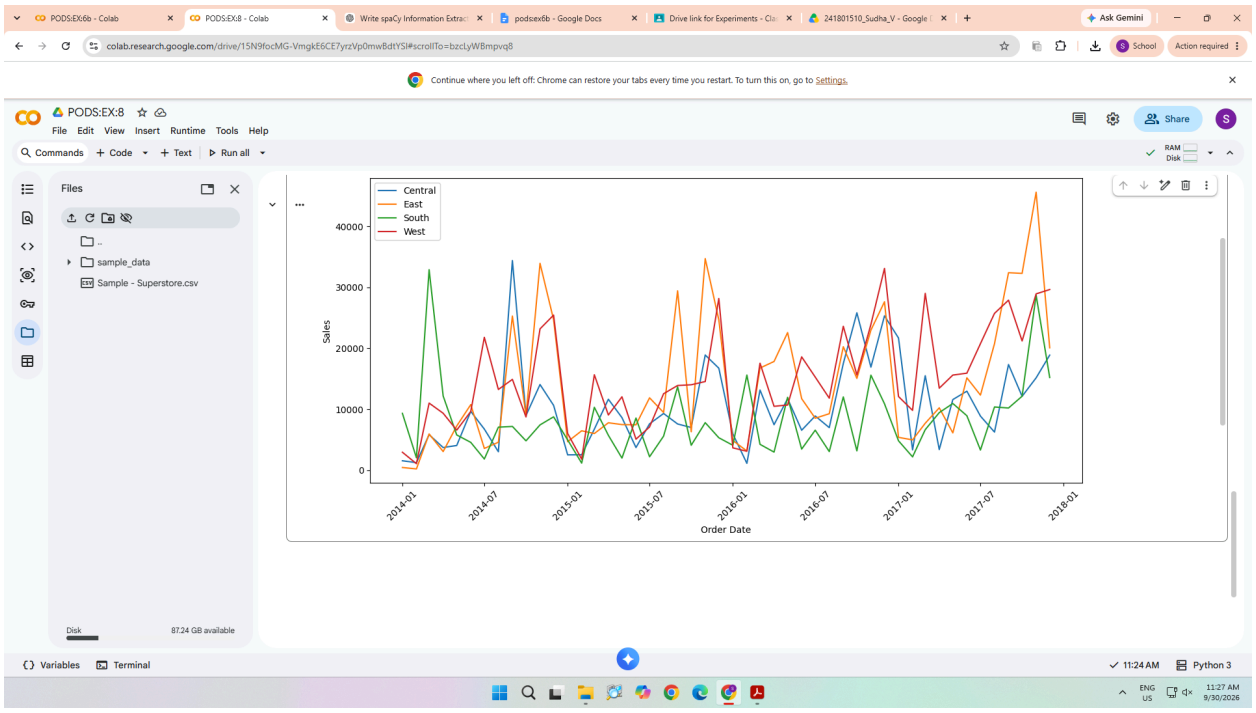
data = df.groupby(["Sub-Category", "Category"])[["Sales", "Profit"]].sum().reset_index()

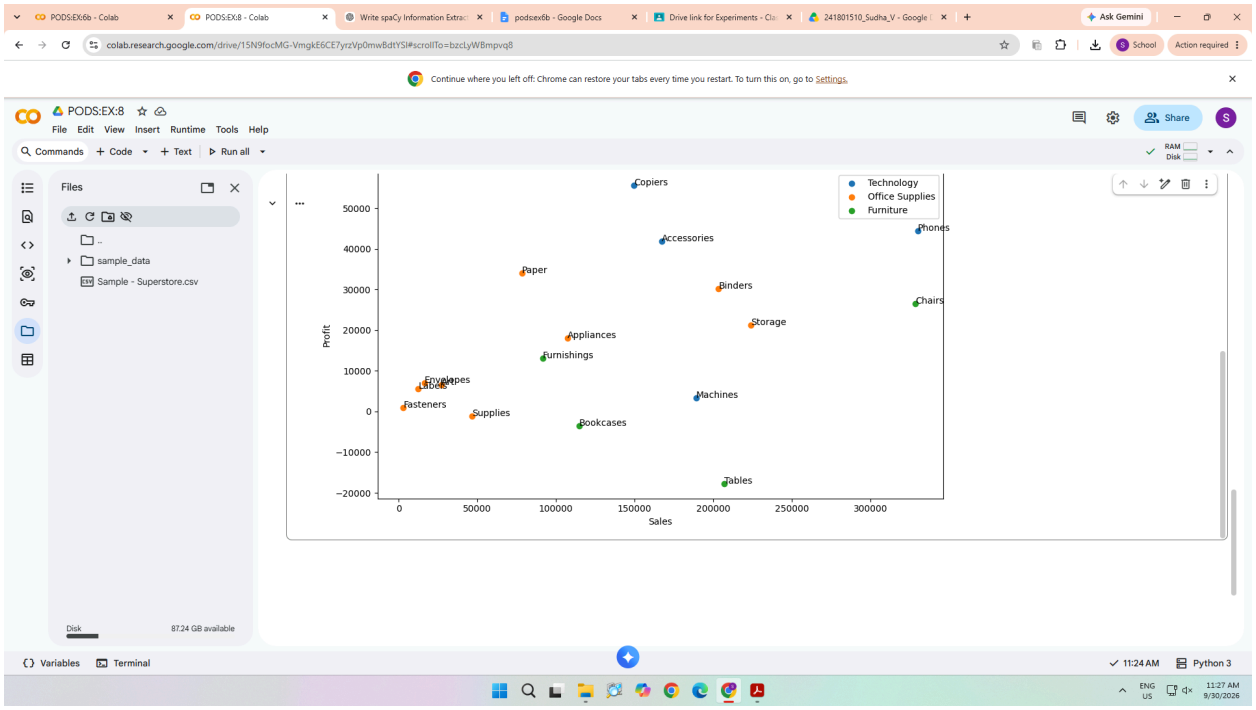
for category, d in data.groupby("Category"):
    plt.scatter(d["Sales"], d["Profit"], label=category)

plt.title("Profit vs Sales by Sub-Category")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.legend()
plt.show()
```

OUTPUT:


plt.xlabel("Sales")
plt.ylabel("Profit")
plt.legend()
plt.tight_layout()
plt.show()
Sub-Category 0
Product Name 0
Sales 0
Quantity 0
Discount 0
Profit 0
dtype: int64
Categories:
['Furniture' 'Office Supplies' 'Technology']
Regions:
['South' 'West' 'Central' 'East']
Sales by Category
Sales
800000
700000
600000
500000
400000
300000
200000
Files
sample_data
Sample - Superstore.csv
Disk
87.24 GB available
Variables
Terminal
11:24 AM
Python 3


Central
East
South
West
Sales
40000
30000
20000
10000
0
2014-01
2014-07
2015-01
2015-07
2016-01
2016-07
2017-01
2017-07
2018-01
Order Date
Files
sample_data
Sample - Superstore.csv
Disk
87.24 GB available
Variables
Terminal
11:24 AM
Python 3



LEARNING OUTCOME:

- Create interactive visualizations using Tableau.
- Analyze sales and profit data.
- Create bar, line, and scatter charts.
- Apply filters to dashboard visualizations.
- Create interactive dashboard actions.
- Combine multiple worksheets into a single dashboard.
- Design an effective and user-friendly business dashboard.
- Publish or export Tableau dashboards for further analysis.